

Optimization – 3

The Euler–Lagrange equation and the direct method

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Theoretical minimum.

Theorem 1 (Euler–Lagrange). If y minimises $J(y) = \int_a^b F(x, y, y') dx$ among C^1 functions with fixed endpoints, then $\frac{d}{dx} F_{y'} = F_y$ on $[a, b]$. If F does not depend on y , then $F_{y'}$ is constant; if F does not depend on x , then $F - y' F_{y'}$ is constant, which is Beltrami’s first integral. If an endpoint is free, the natural boundary condition $F_{y'} = 0$ holds there.

Theorem 2 (constraints and sufficiency). For a minimum of J subject to $K(y) = \text{const}$ the function y satisfies the Euler–Lagrange equation of $J - \lambda K$ for some real λ . Conversely, if $F(x, y, p)$ is convex in (y, p) , then every solution of the Euler–Lagrange equation with the given boundary values is a global minimiser, and it is unique when the convexity is strict.

Fact 1 (completing the square beats the equation). For a quadratic functional it is faster to guess the extremal y_0 and write $J(y) = J(y_0) + \int (\text{something})^2$, which simultaneously proves minimality and uniqueness and needs no smoothness of the competitor. Existence, however, is a separate matter: the direct method needs a minimising sequence, compactness and lower semicontinuity, and without them the infimum may fail to be attained.

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1. Derive the Euler–Lagrange equation, both first integrals and the natural boundary condition. Then solve $\min \int_0^1 (y')^2 dx$ with $y(0) = 0$, $y(1) = 1$ twice: through the equation and by completing the square, and compare what each proof gives.
 2. (RMO) Let F be the set of differentiable functions $f: [0, 1] \rightarrow \mathbb{R}$ with continuous derivative and $f(0) = 0$, $f(1) = 1$. Find $\min_{f \in F} \int_0^1 \sqrt{1 + x^2} (f'(x))^2 dx$ and all functions attaining it.

3. (SEEMOUS) Find the maximum of $\int_0^1 \frac{|f'(x)|^2 |f(x)|}{\sqrt{x}} dx$ over all continuously differentiable $f: [0, 1] \rightarrow \mathbb{R}$ with $f(0) = 0$ and $\int_0^1 |f'(x)|^2 dx \leq 1$.
4. (RMO) Let C be the set of twice differentiable $f: [0, 1] \rightarrow \mathbb{R}$ having at least two zeros, counted with multiplicity, and satisfying $|f''(x)| \leq 1$ everywhere. Find the greatest value of $\int_0^1 |f(x)| dx$ over $f \in C$ and the functions achieving it.
5. Solve by the Euler–Lagrange equation the three classical problems: the brachistochrone, the shape of a hanging chain and the surface of revolution of least area. Point out exactly where Beltrami’s first integral is what makes the computation possible.